

| Question |     |       |  | Marking details                                                                                                                                                                                                                                             | Marks available |          |          |          |          |          |
|----------|-----|-------|--|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|----------|----------|----------|----------|----------|
|          |     |       |  |                                                                                                                                                                                                                                                             | AO1             | AO2      | AO3      | Total    | Maths    | Prac     |
| 3        | (a) | (i)   |  | Sun/ sunlight/ solar                                                                                                                                                                                                                                        | 1               |          |          | 1        |          |          |
|          |     | (ii)  |  | Producer(s)<br>Ignore green plants                                                                                                                                                                                                                          | 1               |          |          | 1        |          |          |
|          |     | (iii) |  | Consumer(s)<br>Reject primary/ secondary/ tertiary                                                                                                                                                                                                          | 1               |          |          | 1        |          |          |
|          | (b) | (i)   |  | TL 5 = 0.01 (%)<br>TL 4 = 0.1 (%)<br>TL 3 = 1(.0) (%)<br><br>All correct = 2 marks<br>Two correct = 1 mark<br>One or none correct = 0 marks                                                                                                                 |                 | 2        |          | 2        | 2        |          |
|          |     | (ii)  |  | Because there is {insufficient/ not enough} <u>energy</u> (to transfer to a TL 6)<br>Reject No energy                                                                                                                                                       |                 |          | 1        | 1        |          |          |
|          |     | (iii) |  | Waste materials/ respiration/ heat/ excretion/ egestion/ not all {eaten/ digested}<br>(Do not accept repair or maintenance and growth of cells)                                                                                                             | 1               |          |          | 1        |          |          |
|          | (c) |       |  | size of the organisms decreases and their numbers increase<br>size of the organisms decreases and their numbers decrease<br>size of the organisms increases and their numbers increase<br><u>size of the organisms increases and their numbers decrease</u> |                 | 1        |          | 1        |          |          |
|          |     |       |  | <b>Question 3 total</b>                                                                                                                                                                                                                                     | <b>4</b>        | <b>3</b> | <b>1</b> | <b>8</b> | <b>2</b> | <b>0</b> |